

1. Administrative & Study Identification

Full Study Title: A Phase 1, Multicenter, Open-label Study to Evaluate the Safety and Preliminary Efficacy of Arlocabtagene Autoleucel (BMS-986393) in Novel Combinations in Participants With Relapsed and/or Refractory Multiple Myeloma and Determine the Recommended Dose for Each Add-on Investigational Component
Short Title/Acronym: A Study to Evaluate the Safety, Effectiveness and Tolerable Dose of Arlocabtagene Autoleucel (BMS-986393) in Novel Combinations in Participants With Relapsed and/or Refractory Multiple Myeloma
Protocol Number: CA088-1005
NCT Number: NCT06121843
Sponsor Name: Bristol Myers Squibb
Investigational Product: Arlocabtagene autoleucel (BMS-986393) (CAR T cells) in combination with alnuctamab, mezigdomide, iberdomide, or elranatamab (Trade Name: Elrexfio)
Phase of Development: PHASE1
Trial Status: RECRUITING (Currently recruiting participants)
Medical Monitor: Bristol Myers Squibb Clinical Operations Lead / Clinical Trials Navigator

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To establish a safe and tolerable dose of arlocabtagene autoleucel (BMS-986393) in combinations with alnuctamab, mezigdomide, iberdomide, and elranatamab in participants with relapsed and/or refractory multiple myeloma (RRMM).
Secondary Objectives: To evaluate the preliminary efficacy of arlocabtagene autoleucel in these novel combinations and determine the recommended dose for each add-on investigational component.

2.2 Background & Rationale To address the critical unmet medical need for effective treatments in patients with relapsed and/or refractory multiple myeloma (RRMM). Despite advances, RRMM remains largely incurable. This trial aims to enhance anti-myeloma activity by combining BCMA-directed CAR T-cell therapy (arlocabtagene autoleucel) with bispecific T-cell engagers (BiTEs) like alnuctamab and elranatamab, or CELMoD agents like iberdomide and mezigdomide.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional
Control Type: Single-arm (Sequential Assignment)
Blinding: Open-label (Masking: None)
Randomization: Non-Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 117 evaluable participants **Number of Sites:** Approximately 17 locations across the United States **Estimated Study Start:** 22-Feb-2024 **Estimated Primary Completion:** 01-Aug-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. History of relapsed and/or refractory multiple myeloma (RRMM) treated with at least 3 (Part 1 and Part 2) or at least 1 but not greater than 3 prior anti-myeloma treatment regimens (Part 2). 3. Measurable multiple myeloma (MM). 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0-1.

4.2 Exclusion Criteria 1. Prior treatment with alnuctamab (Arm A), mezigdomide (Arm B), iberdomide (Arm C), elranatamab (Arm D) or BCMA-targeting therapy (Part 2 Arms A and D). 2. Prior treatment with GPRC5D-targeting therapies. 3. Other protocol-defined inclusion/exclusion criteria that impact patient safety.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Treatment comprises apheresis, lymphodepleting chemotherapy, followed by a single infusion of arlocabtagene autoleucel (BMS-986393). Patients are assigned to Group 1 (+alnuctamab), Group 2 (+elranatamab), Group 3 (+iberdomide), or Group 4 (+mezigdomide). **Route of Administration:** Intravenous (IV) infusion for CAR T cells.

Duration of Treatment: Single infusion with close monitoring for 2 years, followed by long-term health tracking for 15 years.

Regulatory/Reference Documentation: EMA URL for elranatamab (Elrexio) Product Information - https://www.ema.europa.eu/en/documents/product-information/elrexio-epar-product-information_en.pdf

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care pre-conditioning chemotherapy and therapies required for the management of cytokine release syndrome (CRS) or immune effector cell-associated neurotoxicity syndrome (ICANS). **Prohibited:** Prior BCMA-targeting or GPRC5D-targeting therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans, Apheresis
Baseline	Day -5 to Day 0	Pre-conditioning chemotherapy (lymphodepletion)
Intervention	Day 0	Arlocabtagene Autoleucel IV Infusion
Post-Infusion	Weeks 1 - 4	In-patient/intensive out-patient safety monitoring for DLTs, CRS, and ICANS
Follow-Up	Up to	

2 Years | Biopsies, scans, preliminary efficacy assessment (ORR, PFS) | | **Long-Term** | Up to 15 Years | Annual health checks and safety monitoring (FDA mandate for cellular therapies) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Incidence of adverse events (AEs) 2. Incidence of serious adverse events (SAEs) 3. Incidence of adverse events of special interest (AESI) **Measurement Method:** Clinical assessment graded per standard toxicity criteria during the primary evaluation period to establish safe and tolerable dosing.

7.2 Secondary & Exploratory Endpoints Preliminary Efficacy Endpoints: 1. Overall response rate (ORR) 2. Complete response rate (CRR) 3. Very good partial response rate (VGPR) **Pharmacokinetics:** Assessment of CAR T-cell expansion and persistence in peripheral blood.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical evaluations, vital signs, and laboratory assessments, specifically watching for CRS and ICANS. **Serious Adverse Events (SAEs):** Reported standardly within 24 hours. Participants are closely followed for 2 years for major events. **Data Monitoring Committee (DMC):** Internal safety review committees evaluate sequential assignment and dose-escalation safety thresholds.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving any study treatment) and Efficacy Evaluable Set (all participants with measurable disease at baseline who receive the CAR T-cell infusion). **Sample Size Justification:** Target enrollment of \$N=117\$ is based on standard Phase 1 dose-escalation guidelines to establish safety profiles. **Handling of Missing Data:** Standard handling for early phase oncology trials; patients discontinuing prior to DLT evaluation for non-drug related reasons may be replaced.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring during the initial infusion period, transitioning to periodic visits. **Audit**

Trail: Robust electronic data capture (EDC) systems and eCRF locks are maintained.

11. Study Identifiers & Governance

Primary ID: CA088-1005 **Registry Number:** NCT06121843 **Posting ID:** 208446140556121609 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate the Safety, Effectiveness and Tolerable Dose of Arlocabtagene Autoleucel (BMS-986393) in Novel Combinations in Participants With Relapsed and/or Refractory Multiple Myeloma **Official Regulatory Title:** A Phase 1, Multicenter, Open-label Study to Evaluate the Safety and Preliminary Efficacy of Arlocabtagene Autoleucel (BMS-986393) in Novel Combinations in Participants With Relapsed and/or Refractory Multiple Myeloma and Determine the Recommended Dose for Each Add-on Investigational Component

12. Clinical Framework (Multiple Myeloma Specific)

Primary Condition / Preferred UMLS Name: Multiple Myeloma **Disease Areas:** Multiple Myeloma **Semantic Type:** Neoplastic Process **Clinical Definition:** A bone marrow-based plasma cell neoplasm characterized by a serum monoclonal protein and skeletal destruction with osteolytic lesions, pathological fractures, bone pain, hypercalcemia, and anemia. Clinical variants include non-secretory myeloma, smoldering myeloma, indolent myeloma, and plasma cell leukemia. (WHO, 2001) **Disease Classifications:** * **MeSH Code:** D009101 * **HPO Code:** HP:0006775 * **SNOMED ID:** 109989006 **Medical Methodology:** BCMA-directed CAR T-Cell therapy combined with BiTEs or CELMoDs **Optimization Target:** Safe and tolerable maximum dose determination

13. Study Procedures & Methodology

Management Pathway: Apheresis, T-cell engineering, lymphodepletion, CAR-T infusion, and continuous combination therapy **Education Protocol (HCP):** Site training for CAR-T administration, handling CRS (Cytokine Release Syndrome) and ICANS (neurotoxicity) **Education Protocol (Patient):** Onsite training regarding CAR-T cell manufacturing timelines, in-patient requirements, and rigorous 15-year long-term follow-up expectations **Quality Audit Mechanism:** Safety and tolerability tracking, sequential dose-escalation review gates

14. Observation & Follow-Up Schedule

Total Duration: Up to 15 years (2 years of active protocol monitoring, 13 years of safety follow-up) **Onsite Visit Cadence:**

Intensive monitoring in Weeks 1-4, followed by regular interval biopsies/scans **Remote Monitoring Frequency:** Transitioning to annual follow-ups post 2-year mark **Checklist Review Interval:** Regular disease assessment intervals as per IMWG response criteria

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					